

Predicting Loss Curves of LLM Pretraining

Cosine-to-WSD transfer with reproduced scaling-law baselines and two lightweight corrections

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Repository: <https://github.com/Xiaoxiao-2002/TDLT2026-final-project.git>

1.492%

best WSD MAPE after our step-
progress Tissue correction

Project Requirements Covered

Reproduction

Implement Tissue / momentum law and Multi-Power Law under the same cosine-to-WSD setting.

3

learning-rate schedules: cosine, WSD, 8-1-1

Diagnosis

Use residuals and schedule-wise prediction plots to locate baseline failure modes.

2

reproduced baselines: Tissue and MPL

Method

Compare two lightweight Tissue corrections fitted on cosine and tested on WSD and 8-1-1.

2

interpretable Tissue correction variants

Motivation

Pretraining loss curves are often the earliest reliable signal of whether a run is healthy.

- Full LLM pretraining runs are expensive.
- Learning-rate schedules change the entire optimization trajectory.
- A transferable loss law lets us estimate a new schedule before running a full pretraining experiment.

Central question

Can a loss law fitted only on a cosine schedule predict the loss curve under WSD?

The project is about cross-schedule transfer, not only in-schedule curve fitting.

Data and Evaluation Protocol

Fit

Course pkl curve with cosine learning-rate schedule.



Predict

WSD is the required target schedule; 8-1-1 is an extra stress test.



Evaluate

Use log-space R^2 , MAPE, MAE, RMSE, PredE, and residual plots.

input: learning-rate history

output: predicted loss curve

train: cosine

test: WSD + 8-1-1

Related Methods We Reproduce

Tissue / Momentum Law

Compact and interpretable law based on learning-rate accumulated features.

Strength: stable fitting and readable parameters.

Risk: too rigid for the early-stage loss drop.

Multi-Power Law

More expressive: seven-parameter model for loss prediction across schedules.

Strength: richer functional form.

Risk: harder to fit and less directly interpretable.

Our design choice: keep Tissue's interpretability, then test whether small residual corrections can improve schedule transfer.

Baseline 1: Tissue / Momentum Law

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t)$$

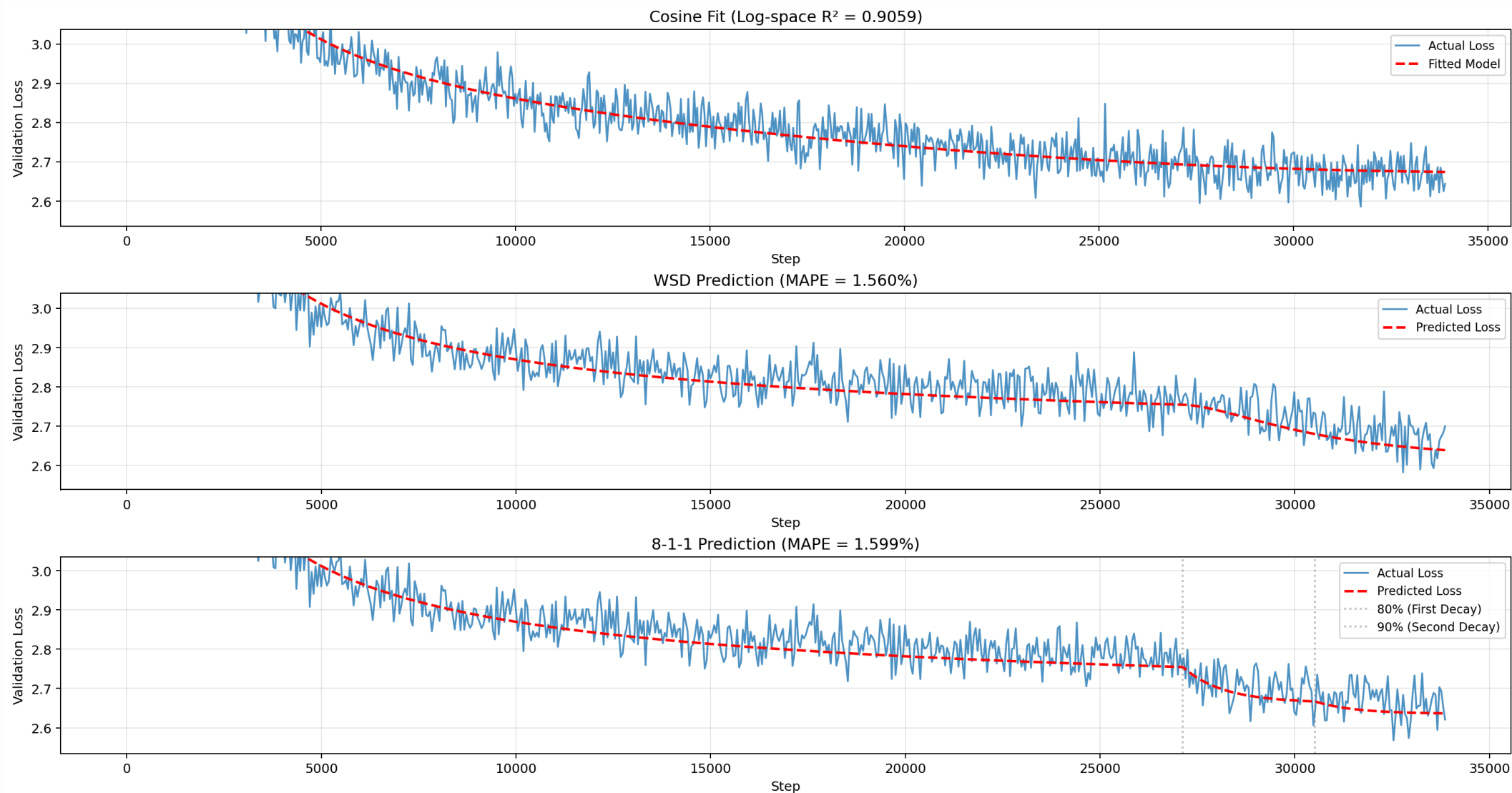
- Fit parameters on cosine.
- Predict WSD and 8-1-1 without refitting.
- Main metric for comparison: WSD MAPE.

1.560%

WSD MAPE for plain Tissue baseline

Observed issue: Tissue fits the general trend, but its residuals reveal biased predictions.

Tissue Baseline: Fit and Transfer



Cosine fit, WSD transfer, and 8-1-1 transfer from the saved Tissue baseline run.

Baseline 2: Multi-Power Law

$$L(t) = L_0 + AS_1(t)^{-\alpha} + B \cdot \text{drop}(t)$$

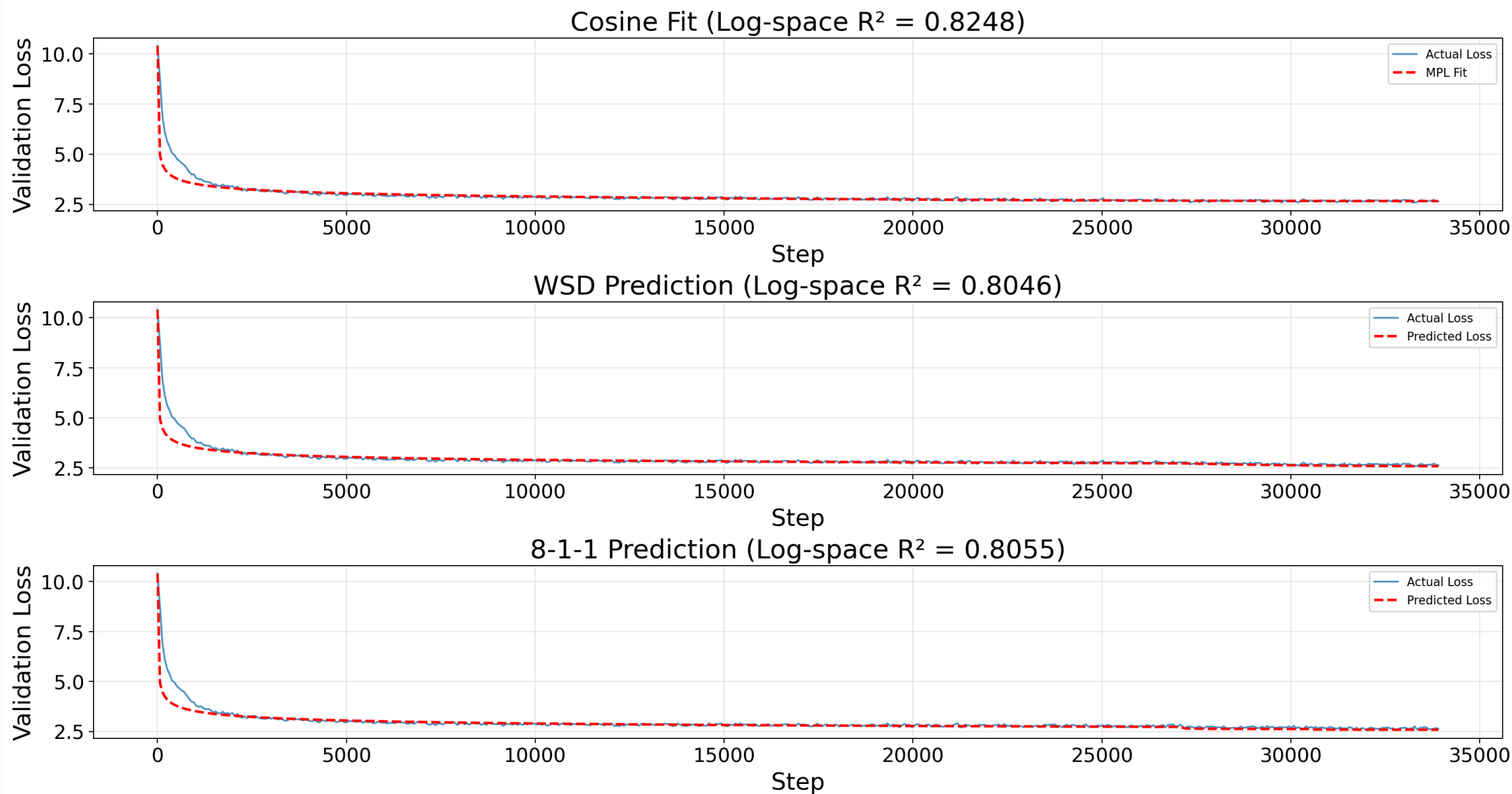
$$\text{drop}(t) = \sum_{i=1}^t (\eta_i - \eta_{i-1}) \left[1 - \left(1 + c\eta_i^{-\gamma} (S_1(t) - S_1(i)) \right)^{-\beta} \right].$$

- Full pkl-based reproduction in `reproduce_2.py`.
- Downsamples the dense pkl curve before fitting, similar to a sparse-checkpoint setup.
- Serves as the more expressive reproduced contrast baseline.

0.805

average test log-space R^2 for MPL

Multi-Power Law: Fit and Transfer

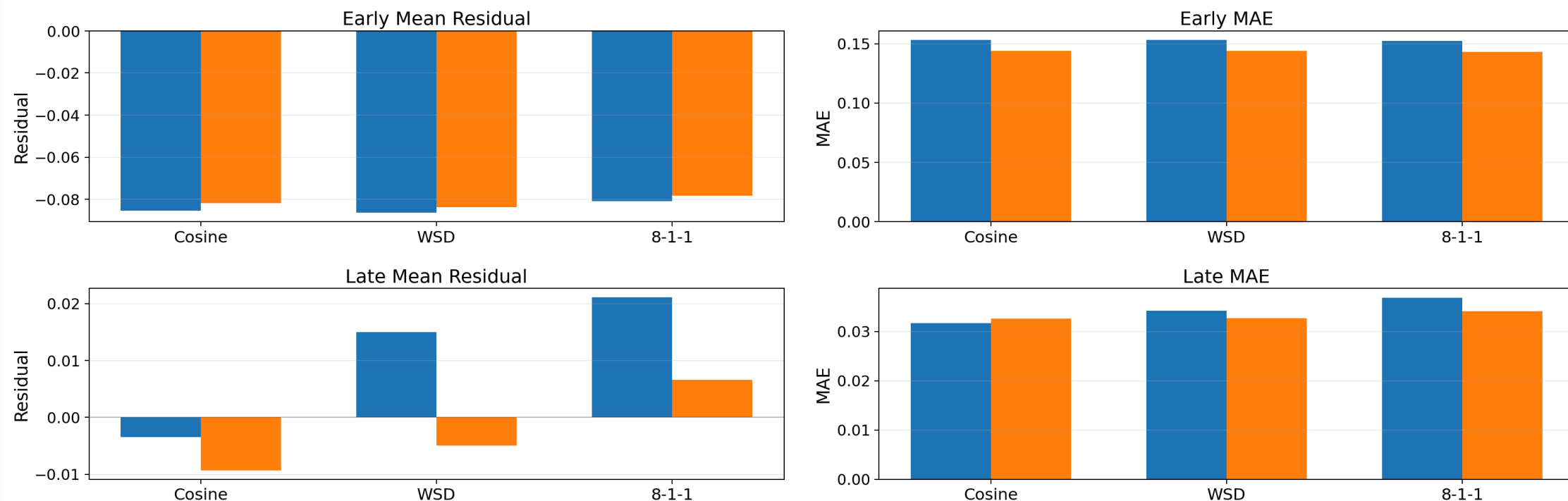


MPL is more expressive, but the pkl-based reproduction is not stronger than the corrected Tissue variant on the chosen metric.

Limitation Analysis: Where Tissue Fails

Residual Pattern Diagnosis for Tissue Baseline vs Improved Variant

Blue: Tissue baseline | Orange: Tissue + progress correction



Negative residuals mean actual loss is lower than predicted loss; the diagnostic shows where the correction moves bias closer to zero.

Method Development: Two Tissue Corrections

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t) - De^{-\tau p(t)}$$

Step-progress correction

Use raw normalized progress: $p(t)=t/T$.

This directly targets the observed early transient bias and is strongest at the beginning.

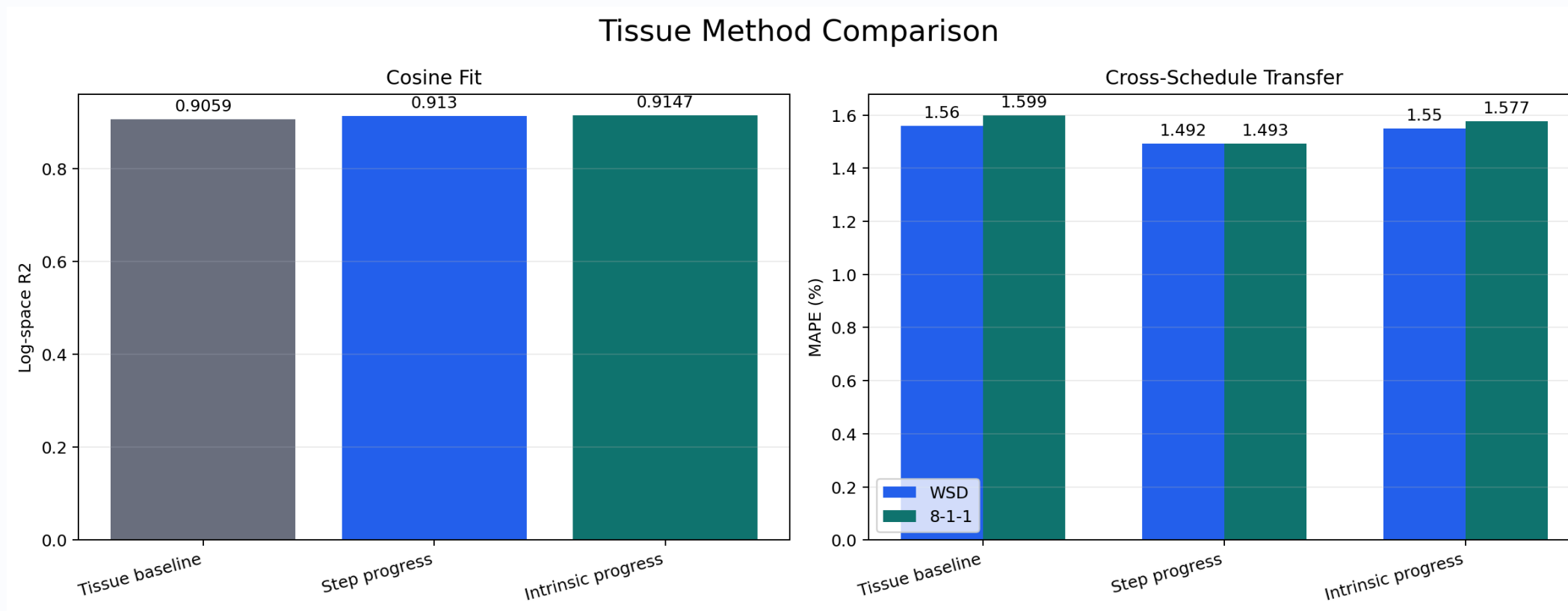
Intrinsic-progress correction

Use learning-rate area progress: $p(t)=S1(t)/S1(T)$.

This is more schedule-aware and follows the intrinsic-time view that effective progress is accumulated step size.

Design constraint: keep the same two added parameters D and τ , so the comparison isolates the progress variable rather than model capacity.

Comparing the Two Corrections



Intrinsic progress fits cosine slightly better, but step progress transfers better to WSD and 8-1-1.

Why Do They Behave Differently?

Step progress wins on transfer

The main failure mode is an early transient overprediction shared by cosine, WSD, and 8-1-1. A simple t/T decay acts like a schedule-agnostic early-bias correction.

Intrinsic progress wins on fit

$S1(t)/S1(T)$ better tracks effective optimization progress inside the cosine training curve, so it improves in-schedule R2.

Interpretation: a theoretically cleaner progress variable can over-adapt to the training schedule when only one fitted curve is available. We therefore report step progress as the main method and intrinsic progress as an ablation.

Main Quantitative Results

Method	Fit metric	Cross-schedule metric	Role
Tissue baseline	cosine log-space $R^2 = 0.905891$	WSD MAPE = 1.5601% 8-1-1 MAPE = 1.5994%	reproduced baseline
Tissue + step-progress correction	cosine log-space $R^2 = 0.912989$	WSD MAPE = 1.4924% 8-1-1 MAPE = 1.4925%	main method
Tissue + intrinsic-progress correction	cosine log-space $R^2 = 0.914654$	WSD MAPE = 1.5504% 8-1-1 MAPE = 1.5766%	ablation
Multi-Power Law	train log-space $R^2 = 0.824753$	test log-space $R^2 = 0.805011$	reproduced contrast

-4.3%

relative WSD MAPE reduction vs Tissue baseline

-6.7%

relative 8-1-1 MAPE reduction vs Tissue baseline

simple

both corrections add only D and τ

Reproducibility and Code Map

reproduce_1.py

Tissue baseline, step-progress correction, intrinsic-progress ablation, residual analysis, and comparison figure.

experiments/

Saved configs, fitted parameters, predictions, residuals, and metrics for each run.

reproduce_2.py

Multi-Power Law reproduction with pkl loading, sampled checkpoints, metrics, and plot export.

figures/

Main visual evidence used in the final slides.

Division of Labor

Tianze Lu

Main code implementation,
Tissue baseline reproduction,
improved method experiments.

Ruoyu Wang

Final slide design, result
organization and polishing.

Feiyue Ye

Literature notes, result checking,
and final formatting support.

Source code, saved experiment artifacts, figures, and final slides are included in the repository.

Conclusions and Next Steps

- Cross-schedule loss prediction is feasible, but schedule-sensitive.
- Tissue is interpretable but misses the early-stage loss drop.
- MPL is a useful reproduced contrast baseline.
- Step-progress correction gives the best transfer result.
- Intrinsic-progress correction gives a cleaner theory-motivated ablation but weaker transfer.

Final takeaway

Residual diagnosis can guide small, interpretable improvements, but transfer metrics should decide the final method.

Next: fit multiple schedules or regularize schedule-aware residual terms to avoid over-adapting to cosine.